NYU|Tandon

MA 2034 WORKSHEET 2 FALL 2026

Alligator Population Dynamics, Harvesting, and Lake Mixing

Name: _____ Net ID: _____

Instructions. Show your work and explain your reasoning when requested. Use phase lines to justify long-term behavior. Write your answers on this worksheet. **You must align your pages with the outline on Gradescope–otherwise your work will not be graded.**

Introduction: Alligator Population Dynamics

The American alligator is one of the best-known examples of a wildlife population that recovered after severe decline. During much of the twentieth century, alligator populations were reduced by hunting and loss of habitat. Legal protection and conservation measures allowed populations to recover, and in many areas controlled harvesting is now permitted.

Wildlife managers therefore face a mathematical question: *How many animals can be removed from a population each year without causing the population to collapse?* A population that appears large may still be vulnerable if harvesting exceeds its natural ability to replace the animals removed. The answer also depends on the current population size, since population growth is not equally rapid at all population levels.

In this worksheet, we use a logistic differential equation to model an alligator population and then introduce harvesting. We will investigate equilibrium populations, stability, thresholds, and a bifurcation that occurs when harvesting becomes too large. The purpose is not only to solve the differential equations, but also to interpret what the mathematics says about managing a biological population.

Let $P(t)$ represent the number of alligators in a managed population t years after observations begin. In the absence of harvesting, assume

$$\boxed{\frac{dP}{dt} = 0.75P\left(1 - \frac{P}{1200}\right)}.$$

Thus $r = 0.75$ yr^{-1} and the carrying capacity is $K = 1200$.

Part I. Understanding the Logistic Model (12 Points)

1. Explain in words what $r = 0.75$ and $K = 1200$ mean.

2. Without solving the differential equation, describe the long-term behavior if $P(0) = 300$.

3. Without solving the differential equation, describe the long-term behavior if $P(0) = 1400$?

4. At what population would you expect the natural population growth to be greatest? Explain.

Part II. Constant Harvesting (36 Points)

Suppose wildlife managers allow h alligators to be harvested each year. The model becomes

$$\boxed{\frac{dP}{dt} = 0.75P\left(1 - \frac{P}{1200}\right) - h}.$$

1. Equilibria

a. Set $dP/dt = 0$ and show that the equilibrium equation can be written

$$P^2 - 1200P + 1600h = 0.$$

b. Use the quadratic formula to show that

$$\boxed{P = \frac{1200 \pm \sqrt{1,440,000 - 6400h}}{2}}.$$

c. What must be true about h for two distinct equilibrium populations to exist?

2. Harvesting 45 Alligators per Year

Let $h = 45$.

a. Find the two equilibrium populations.

b. Draw a phase line and indicate the direction of solutions.

c. Classify each equilibrium as stable or unstable.

d. What happens in the long run if $P(0) = 500$?

e. What happens if $P(0) = 50$?

f. Explain the biological importance of the smaller equilibrium.

3. Harvesting 135 Alligators per Year

Let $h = 135$.

a. Find the equilibrium populations.

b. Draw and label a phase line.

c. Classify each equilibrium.

d. Predict the long-term behavior if $P(0) = 600$.

e. Predict what happens if $P(0) = 200$.

f. Compare the equilibria for $h = 45$ and $h = 135$. What happens to the stable and unstable equilibria as harvesting increases?

Part III. Maximum Sustainable Harvest (14 Points)

The natural population growth is

$$G(P) = 0.75P\left(1 - \frac{P}{1200}\right).$$

1. At what population is $G(P)$ maximized?

2. Find the maximum value of $G(P)$.

3. Determine the theoretical maximum sustainable constant harvest $h_{\max}$.

4. Draw a phase line when $h = h_{\max}$.

5. What happens if the initial population is slightly above the critical equilibrium?

6. What happens if it is slightly below it?

7. Why might harvesting at exactly $h_{\max}$ be risky in a real ecosystem?

Part IV. Bifurcation (13 Points)

When relatiavely small changes in initial conditions causes distinctly different solution behaviors, this is called Bifurcation. We will examine the bifurcation diagram for the alligator population that we have been studying in this worksheet. Use

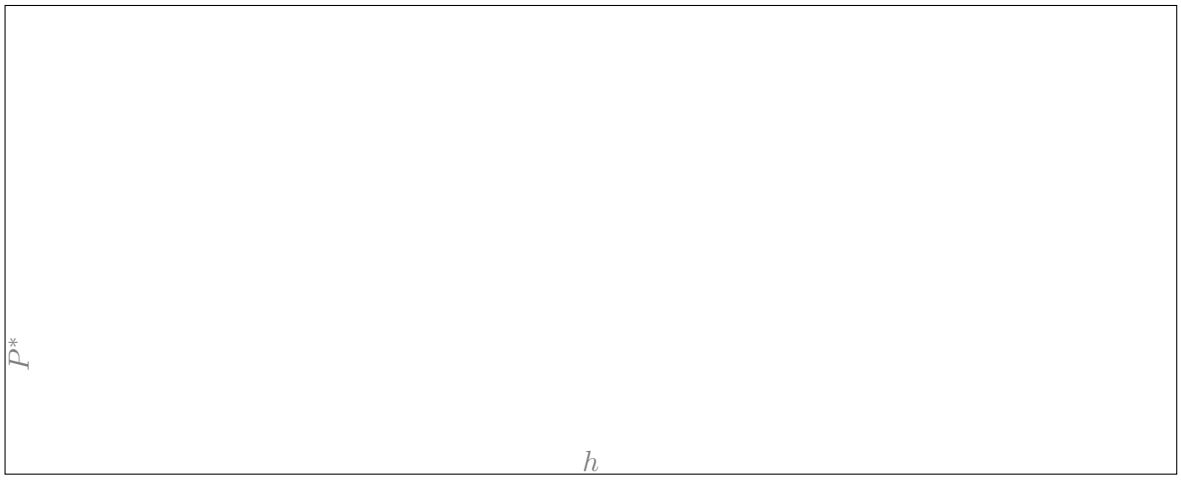
$$P^*(h) = \frac{1200 \pm \sqrt{1,440,000 - 6400h}}{2}$$

to calculate the equilibrium populations for only the following four harvest levels:

$$\boxed{h = 0, \quad 90, \quad 180, \quad 225.}$$

h	Lower equilibrium	Upper equilibrium
0		
90		
180		
225		

Sketch a bifurcation diagram with h on the horizontal axis and equilibrium population P^* on the vertical axis. Use a solid curve for stable equilibria and a dashed curve for unstable equilibria.



a. What happens to the two equilibria as h increases? Where do they meet, and what happens afterward?

Part V. Mixing Problem: Fertilizer Runoff into a Lake (25 Points)

Part V is not related to Parts 1–IV, is is a separate question on Mixing.

A lake near agricultural land has a volume of 500 million gallons. Water containing fertilizer runoff flows into the lake at a rate of 10 million gallons per day. The incoming water contains 3 tons of dissolved fertilizer per million gallons. Assume the lake is well mixed and water leaves at the same rate, so the volume remains constant. Initially the lake contains 200 tons of fertilizer.

Let $Q(t)$ be the amount of fertilizer, in tons, in the lake after t days. Recall

$$\boxed{\text{Rate of change} = \text{Rate in} - \text{Rate out.}}$$

1. How many tons of fertilizer enter the lake each day?

2. What is the concentration of fertilizer in the lake at time t, in tons per million gallons?

3. Determine the rate at which fertilizer leaves the lake.

4. Show that
$$\boxed{\frac{dQ}{dt} = 30 - \frac{Q}{50}}.$$

5. Find the equilibrium amount of fertilizer in the lake.

6. Without solving the differential equation, determine whether $Q(t)$ initially increases or decreases. Explain.

7. Solve the initial-value problem
$$Q' = 30 - \frac{Q}{50}, \qquad Q(0) = 200.$$

8. Determine $\lim_{t\to\infty} Q(t)$ and explain its environmental meaning.

Reducing the Runoff

After environmental regulations are introduced, the fertilizer concentration entering the lake is reduced from 3 to 0.5 tons per million gallons. Assume the lake initially contains the equilibrium amount found above.

a. Write the new differential equation.

b. Find the new equilibrium amount of fertilizer.

c. Without completely solving the new equation, explain what happens after the regulations take effect.

d. Does reducing fertilizer runoff immediately remove the pollution already present? Find $Q(25)$.Explain.

e. How much fertilizer remains after 50 days, after 100 days,after 150 days?

Connecting the Two Models

In 2–3 sentences, explain what an equilibrium means in the alligator model and in the lake-mixing model, and why knowing an equilibrium alone does not necessarily tell us whether an environmental situation is desirable.